

■ CRT LOCK — 2-Speaker Traffic Signal Presentation Script

Format: 2-Speaker Dialogue / Conversation

Characters:

- Alex (Speaker 1): The Everyday Driver / Host (Curious, relatable, asks questions)
- Sam (Speaker 2): The Traffic Tech Lead (Explains concepts simply, runs the live demo)

Target Duration: ~3–4 Minutes

■ Scene 1: The Problem (Stuck in Traffic)

Alex:

> *"Sam, be honest with me... why does it feel like whenever I'm in a rush, every single traffic light turns red right in front of me?"*

Sam:

> *"Haha! It's actually a classic math problem. Traffic signals along a busy street operate on different timer cycles — like 3 minutes, 5 minutes, or 7 minutes. If they aren't synchronized, you get stuck stopping at every block."*

Alex:

> *"So how do city planners fix that? How do you get a 'Green Wave' where a car can drive straight through all green lights?"*

Sam:

> *"That's where a 2,000-year-old math theorem called the Chinese Remainder Theorem comes in. And to prove it, we built this interactive visual app called CRT Lock!"*

■ Scene 2: The Traffic Signal Analogy

Sam: **(Pointing to the app screen at `http://localhost:5173/`)*

> *"Imagine our street has 3 traffic signals:"*

> - **Signal 1 changes every 3 minutes.**

> - **Signal 2 changes every 5 minutes.**

> - **Signal 3 changes every 7 minutes.**

>

> *"In our app, each rotating gear is a traffic signal. The slot in the gear represents a GREEN LIGHT, and this sliding key rod is your car trying to drive through."*

Alex:

> *"Got it! So if all three gear slots line up, the rod slides through — meaning my car hits green lights across the whole street?"*

Sam:

> *"Spot on! But if even ONE signal is red, the gear blocks the rod — CLACK! — and your car is forced to stop."*

■ Scene 3: Live Interactive Demo

Alex:

> *"Alright, let's test it! Suppose Signal A turned green 2 minutes ago, Signal B turned green 3 minutes ago, and Signal C turned green 2 minutes ago. What minute should I dial?"*

Sam:

> *"Let's try a random guess first. Give me a number on the dial."*

Alex:

> *"Let's try Minute 20!"*

Sam: *(Sets main dial to 20)*

> *"Okay! At Minute 20:"*

> - *Signal 1 (3-min timer): GREEN light!*

> - *Signal 2 (5-min timer): RED light!*

> - *Signal 3 (7-min timer): RED light!*

>

> *Let's hit Check Lock to try driving through..."*

> *(Clicks 'Check Lock' — Loud THUD sound as rod hits a dead stop)*

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> *"BAM! Red light at Intersection #2! Your car is stuck."*

Alex:

> *"Ouch! Okay, so Minute 20 doesn't work. How do we find the exact minute when ALL THREE turn green?"*

Sam:

> *"That's what the Chinese Remainder Theorem calculates for us. For our 3, 5, and 7 minute timers, the exact magic moment is Minute 23! Dial it to 23."*

Alex: *(Dials to 23)*

> *"Look! All three signal lights on the dashboard turned GREEN!"*

Sam:

> *"Now let's drive through..."*

> *(Clicks 'Check Lock' — Smooth sliding sound, spring latch pops open!)*

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> *"The rod glides straight through all three gears! Click Open..."*

> *(Clicks 'Open' — Door swings open with victory chime)*

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> *"The door opens revealing 23 — the exact green wave minute!"*

■ Scene 4: Wrap-Up & Everyday Impact

Alex:

> *"That's awesome! So the Chinese Remainder Theorem basically finds the one secret timing where different repeating cycles align?"*

Sam:

> *"Exactly. And beyond traffic lights, engineers use this exact same math for fiber-optic internet cables, computer chip processing, and cybersecurity encryption."*

Alex:

> *"It turns heavy textbook math into a visual puzzle you can actually see and hear. Thanks everyone!"*

■ Speaker Role Assignments Summary